Metadata for Dalry (DAL), Ayrshire, UK

Description of data set	Magnetic time-series from the BGS SWIGS differential magnetometer method (DMM) systems. Funded by NERC, grant number: NE/P017231/1 "Space Weather Impact on Ground-based Systems (SWIGS)". These data consist of measurements of the Earth's natural magnetic field at the remote site (DALR) and the natural magnetic field plus the field created by GIC at the underline site (DALU). The database will include .xyz files with the DMM data and one document with metadata. See Hübert et al. (2020) for further details.
Location (in lat/long): underline station	55.7057862 N, -4.7912722 E
Location (in lat/long): remote station	55.7040588 N, -4.7882232 E
Distance between DMM sites	277 m
Description of site (i.e. open moorland)	Open fields within a new tree plantation
Equipment used	Sensys 3DFM fluxgate magnetometer and Earth Data digitisers (EDR 209, 24bit)
Installation date	28/05/2019
Removal date	03/06/2021
Number of days collected	DALR: 630, DALU: 526
Data sampling rate	1 Hz
Notes of known data gaps	DALU: Data gaps: 28/05-18/06/2019, 01/08/2019-23/10/2019, 28/12/2019-11/01/2020, 27/12/2020-05/01/2021. DALR: Data gaps: 01/01/2020-06/01/2020, 08/10/2020-16/11/2020, 27/12/2020-05/01/2021.
Qualitative estimate of site quality	Very good
Orientation angle of HV wires relative to true north	89° (i.e. power lines run approximately E-W)
Alignment of magnetometers	Magnetic north as of 28/05/2019
Voltage level of the lines	400 kV (Strathaven – Hunterston)
Height above the ground of each wire (m)	Lower: 10.45 m, Middle: 18.35 m, Upper: 25.25 m
Underline magnetometer	Placed under southern circuit, 18m to second circuit
File Type	ASCII text (.xyz)
File format	YYYY-MM-DD HH:MM:SS X Y Z Five columns with 86400 rows (plus one header row). Magnetic values in units of nT (10^-9 T). Empty values denoted by: 999999.00
Citation	Hübert, J., Beggan, C. D., Richardson, G. S., Martyn, T., & Thomson, A. W. P. (2020). Differential magnetometer measurements of geomagnetically induced currents in a complex high voltage network. Space Weather, 18, e2019SW002421. doi: 10.1029/2019SW002421